D1 Mensuration — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

D1_Q1: Perimeter of 2D Shapes

Hook

Imagine you're building a fence around your garden. The total length of fencing you need is the perimeter — the distance around the outside of the garden. Every day, architects, engineers, and designers calculate perimeters for everything from picture frames to running tracks.

Prerequisites

- Understanding of basic shapes (square, rectangle, triangle, circle)
- Addition and multiplication skills
- Knowledge of pi as approximately 3.142

Core Explanation

The **perimeter** is the total distance around the edge of a 2D shape. For polygons, it is simply the sum of all side lengths.

Perimeter Formulas

Shape	Formula
Square (side s)	P = 4s
Rectangle (length l, width w)	P = 2(l + w)
Triangle (sides a, b, c)	P = a + b + c
Regular polygon (n sides, side s)	P = n x s
Circle (radius r)	C = 2 x pi x r
Semicircle (radius r)	Arc = pi x r, Total = pi x r + 2r

Circumference of a Circle
The perimeter of a circle is called its **circumference**. The formula is C = 2 x pi x r, where r is the radius. Alternatively, C = pi x d, where d = 2r is the diameter. Use pi = 3.142 unless the question specifies otherwise.

Finding Missing Sides
When given the perimeter and some side lengths, subtract the known sides from the total perimeter to find missing ones.

Word Problems
Always identify which sides contribute to the perimeter. For compound shapes, trace the outer boundary — do not count interior edges.

Worked Example

Problem: A rectangle has length 12 cm and width 8 cm. Find its perimeter.

Step 1: Write the formula. P = 2(l + w)

Step 2: Substitute. P = 2(12 + 8) = 2(20)

Step 3: Calculate. P = 40 cm

Answer: The perimeter is 40 cm.

Problem: A circular running track has a diameter of 70 m. Find its circumference (use pi = 22/7).

Step 1: Write the formula. C = pi x d

Step 2: Substitute. C = (22/7) x 70 = 22 x 10

Step 3: Calculate. C = 220 m

Answer: The circumference is 220 m.

Common Mistakes

X Forgetting to include all sides when calculating the perimeter of a polygon.

OK Fix: Trace around the shape with your finger, counting each side as you go. Write down every side length be

X Using the diameter instead of radius in C = 2 x pi x r, or vice versa.

OK Fix: Remember: C = 2 x pi x r uses the radius. If given the diameter, halve it first: r = d/2.

X Adding interior lines when finding the perimeter of a compound shape.

OK Fix: Perimeter is the distance around the OUTSIDE only. Ignore lines inside the shape.

Summary

- Perimeter is the total distance around a 2D shape — sum all side lengths for polygons.
- Circumference of a circle: C = 2 x pi x r = pi x d.
- For semicircles, add the arc (pi x r) plus the diameter (2r).
- Find missing sides by subtracting known sides from the total perimeter.
- Use pi = 3.142 unless stated otherwise.

D1_Q2: Area of 2D Shapes

Hook

How many tiles do you need to cover a floor? How much paint for a wall? These questions are about area — the amount of space inside a 2D shape. Area calculations appear constantly in construction, landscaping, and manufacturing.

Prerequisites

- Perimeter concepts
- Basic multiplication and squaring
- Understanding of base and height

Core Explanation

Area measures the amount of space inside a 2D shape, measured in square units (cm^2, m^2, etc.).

Area Formulas

Shape	Formula
Square (side s)	A = s^2
Rectangle (length l, width w)	A = l x w
Triangle (base b, height h)	A = (1/2) x b x h
Parallelogram (base b, height h)	A = b x h
Trapezium (parallel sides a, b, height h)	A = (1/2)(a + b)h
Circle (radius r)	A = pi x r^2
Semicircle (radius r)	A = (1/2) x pi x r^2

Key Points

- Height must be measured perpendicular to the base (at right angles).
- For circles, square the radius first, then multiply by pi.
- Always include the correct square units (cm^2, m^2, mm^2).

Unit Conversions

1 cm^2 = 100 mm^2
1 m^2 = 10,000 cm^2
1 km^2 = 1,000,000 m^2

When converting between square units, remember that each linear conversion is squared: 1 m = 100 cm, so 1 m^2 = 100 x 100 = 10,000 cm^2.

Worked Example

Problem: A triangle has base 10 cm and perpendicular height 6 cm. Find its area.

Step 1: Write the formula. A = (1/2) x b x h

Step 2: Substitute. A = (1/2) x 10 x 6 = 5 x 6

Step 3: Calculate. A = 30 cm^2

Answer: The area is 30 cm^2.

Problem: Find the area of a circle with radius 7 cm (use pi = 22/7).

Step 1: Write the formula. A = pi x r^2

Step 2: Substitute. A = (22/7) x 7 x 7 = 22 x 7

Step 3: Calculate. A = 154 cm^2

Answer: The area is 154 cm^2.

Common Mistakes

X Using the slanted side as the height in a triangle or parallelogram.

OK Fix: Height must be the perpendicular distance from the base to the opposite vertex/side. Look for the right

X Forgetting to square the radius in the circle area formula.

OK Fix: A = pi x r^2 means r x r first, then multiply by pi. Do NOT do (pi x r)^2.

X Confusing area and circumference formulas for circles.

OK Fix: Remember: area involves r^2 (square units), circumference involves r or d (linear units). 'Area is squar

X Incorrect unit conversions — converting 50 cm^2 to 0.5 m^2 instead of 0.005 m^2.

OK Fix: 1 m^2 = 10,000 cm^2, so divide by 10,000 when converting from cm^2 to m^2.

Summary

- Area measures the space inside a 2D shape, in square units.
- Key formulas: rectangle (l x w), triangle (1/2 x b x h), circle (pi x r^2), trapezium (1/2(a+b)h).
- Height must be perpendicular to the base.
- Use pi = 3.142 unless stated. Square the radius before multiplying by pi.
- Convert square units carefully: 1 m^2 = 10,000 cm^2, 1 cm^2 = 100 mm^2.

D1_Q3: Area and Perimeter of Composite Shapes

Hook

Real-world objects aren't just simple rectangles or circles — a swimming pool might be a rectangle with a semicircular end, and a garden might combine several shapes. Composite shapes are everywhere, and calculating their perimeter and area is a key IGCSE skill.

Prerequisites

- Area and perimeter of basic 2D shapes
- Addition and subtraction skills
- Identifying parts of a shape

Core Explanation

A **composite shape** is made up of two or more simple shapes (rectangles, triangles, circles, etc.) joined together.

Strategy for Area

1. **Split** the composite shape into simple shapes.
2. **Calculate** the area of each simple shape.
3. **Add** all the areas together.

Sometimes you need to subtract area (e.g., area of a path around a garden = area of outer rectangle minus area of inner rectangle).

Strategy for Perimeter

1. **Trace** the outer boundary of the shape.
2. **Identify** which sides of the component shapes are on the outside.
3. **Add** only the outside edge lengths.

Common Composite Shapes

- L-shape: two rectangles joined at right angles.
- Rectangle with semicircle on one side.
- Annulus (ring): area = pi(R^2 - r^2) where R is outer radius, r is inner radius.
- Rectangular path around a rectangle.

Finding Missing Lengths
Use the given dimensions to deduce missing lengths. For an L-shape, the total vertical height equals the sum of the vertical segments.

Worked Example

Problem: An L-shaped garden consists of two rectangles. Rectangle A is 8 m by 5 m. Rectangle B is 3 m by 4 m. Find the total area.

Step 1: Split into two rectangles.
Area A = $8 \times 5 = 40$ m^2
Area B = $3 \times 4 = 12$ m^2

Step 2: Add the areas.
Total area = $40 + 12 = 52$ m^2

Answer: The total area is 52 m^2.

Problem: A rectangle of length 10 m and width 6 m has a semicircle of radius 3 m attached to one short side. Find the total area (use pi = 3.142).

Step 1: Calculate rectangle area. $A_{rect} = 10 \times 6 = 60$ m^2

Step 2: Calculate semicircle area. $A_{semi} = (1/2) \times \pi \times r^2 = 0.5 \times 3.142 \times 9 = 14.139$ m^2

Step 3: Add. Total = $60 + 14.139 = 74.139$ m^2

Answer: The total area is approximately 74.14 m^2.

Common Mistakes

X Counting interior edges when finding perimeter.

OK Fix: Perimeter is only the outside boundary. If two rectangles share an edge, that edge is NOT part of the pe

X Adding areas but forgetting to subtract holes (e.g., a path or a window).

OK Fix: For shapes with holes, calculate the outer area first, then subtract the area of the hole(s).

X Incorrectly splitting the shape and double-counting regions.

OK Fix: Draw your split lines clearly. Each part should be a complete simple shape, and the parts should not ov

Summary

- Composite shapes are made of simple shapes joined together.
- For area: split into simple shapes, calculate each area, then add (or subtract for holes).
- For perimeter: trace the outer boundary only — do not include interior edges.
- Use given dimensions to find missing lengths.
- Check your work by estimating a reasonable answer.

D1_Q4: Surface Area of Prisms

Hook

How much cardboard is needed to make a box? How much wrapping paper for a gift? The answer is the surface area — the total area of all the faces of a 3D object. This is essential in manufacturing, packaging, and construction.

Prerequisites

- Area of rectangles, triangles, circles
- Understanding of 3D shapes
- Net diagrams of prisms

Core Explanation

A **prism** is a 3D shape with a uniform cross-section. It has two identical parallel faces (the ends) and rectangular side faces.

Surface Area of a Prism

Surface Area = 2 x Area of cross-section + Perimeter of cross-section x Length

Where the cross-section is the shape of the two ends and the length is the distance between them.

Common Prisms

Prism	Cross-section	Faces
Cuboid (rectangular)	Rectangle	6 rectangular faces
Triangular prism	Triangle	2 triangles + 3 rectangles
Cylinder	Circle	2 circles + 1 curved rectangle
Cube	Square	6 equal squares

Surface Area of a Cylinder

SA = 2 x pi x r^2 + 2 x pi x r x h

Where r is the radius, h is the height.
The curved surface area is 2 x pi x r x h (imagine unrolling the curved part into a rectangle).

Net Diagrams
Drawing the net of a prism helps visualise all faces. The net is the 3D shape laid flat.

Units
Surface area is measured in square units (cm^2, m^2).

Worked Example

Problem: A cuboid has length 5 cm, width 3 cm, and height 4 cm. Find its total surface area.

Step 1: Identify the faces.
Two faces: 5 x 3 = 15 cm^2 each
Two faces: 5 x 4 = 20 cm^2 each
Two faces: 3 x 4 = 12 cm^2 each

Step 2: Calculate total.
SA = 2(15) + 2(20) + 2(12) = 30 + 40 + 24 = 94 cm^2

Answer: The total surface area is 94 cm^2.

Problem: A cylinder has radius 3 cm and height 10 cm. Find its surface area (use pi = 3.142).

Step 1: Area of two circular ends. 2 x pi x r^2 = 2 x 3.142 x 9 = 56.556 cm^2

Step 2: Curved surface area. 2 x pi x r x h = 2 x 3.142 x 3 x 10 = 188.52 cm^2

Step 3: Add. Total = 56.556 + 188.52 = 245.076 cm^2

Answer: The total surface area is approximately 245.08 cm^2.

Common Mistakes

X Forgetting to multiply by 2 for the two identical end faces of a prism.
OK Fix: The formula SA = 2 x (area of cross-section) + (perimeter of cross-section x length) reminds you there

X Using the diameter instead of radius in cylinder calculations.
OK Fix: Always halve the diameter to get the radius. Both end circles and the curved surface need the radius.

X Forgetting the curved surface area of a cylinder.
OK Fix: A cylinder has THREE parts: two circles and the curved rectangle. All three contribute to surface area.

X Confusing surface area with volume.
OK Fix: Surface area is in square units (covering the outside). Volume is in cubic units (filling the inside).

Summary

- Surface area = total area of all faces of a 3D shape.
- For a prism: SA = 2 x (area of cross-section) + (perimeter of cross-section x length).
- For a cylinder: SA = 2 x pi x r^2 + 2 x pi x r x h.
- Use net diagrams to visualise all faces.
- Surface area is in square units; volume is in cubic units.

D1_Q5: Volume of Prisms and Cylinders

Hook

How much water can a swimming pool hold? How much concrete is needed for a pillar? These questions are about volume — the amount of space inside a 3D object. Volume calculations are essential in engineering, construction, and everyday life.

Prerequisites

- Area of 2D shapes
- Understanding of 3D shapes and prisms
- Multiplication skills

Core Explanation

Volume measures the amount of space inside a 3D shape, measured in cubic units (cm^3, m^3, etc.).

Volume of a Prism

For any prism: **Volume = Area of cross-section x Length**

The cross-section is the uniform shape that repeats through the prism.

Specific Formulas

Shape	Formula
Cuboid (l x w x h)	V = l x w x h
Cube (side s)	V = s^3
Triangular prism	V = (1/2 x b x h_triangle) x L
Cylinder	V = pi x r^2 x h

Volume of a Cylinder

V = pi x r^2 x h

Where r is the radius of the circular cross-section and h is the height (or length) of the cylinder.

Unit Conversions for Volume

1 cm^3 = 1,000 mm^3
1 m^3 = 1,000,000 cm^3
1 litre = 1,000 cm^3 = 1,000 mL
1 m^3 = 1,000 litres

Capacity Problems

Sometimes you need to find how much liquid a container can hold. Convert volume to litres or millilitres using the conversions above.

Worked Example

Problem: A cuboid fish tank measures 60 cm by 30 cm by 40 cm. Find its volume.

Step 1: Write the formula. V = l x w x h

Step 2: Substitute. V = 60 x 30 x 40

Step 3: Calculate. V = 72,000 cm^3

Answer: The volume is 72,000 cm^3. This equals 72 litres.

Problem: A cylindrical water tank has radius 50 cm and height 120 cm. Find its volume in litres (use pi = 3.142).

Step 1: V = pi x r^2 x h = 3.142 x 50 x 50 x 120

Step 2: V = 3.142 x 2500 x 120 = 3.142 x 300,000

Step 3: V = 942,600 cm^3

Step 4: Convert to litres. 942,600 / 1000 = 942.6 litres

Answer: The tank holds approximately 942.6 litres.

Common Mistakes

X Using the slanted edge length instead of the perpendicular height in a cylinder or prism.
OK Fix: Volume is area of cross-section times the perpendicular height, not the slanted length.

X Forgetting to square the radius in the cylinder volume formula.
OK Fix: V = pi x r^2 x h means r x r first, then multiply by pi and h.

X Using diameter instead of radius.
OK Fix: Always halve the diameter to get the radius before using it in the formula.

X Confusing volume conversion factors: 1 m^3 = 1,000,000 cm^3 (not 1,000).
OK Fix: Linear: 1 m = 100 cm. Area: 1 m^2 = 10,000 cm^2. Volume: 1 m^3 = 1,000,000 cm^3. Each dimensi

Summary

- Volume of any prism = area of cross-section x length.

- Cuboid: V = l x w x h. Cylinder: V = pi x r^2 x h.
- The cross-section is the shape that repeats through the prism.
- 1 litre = 1,000 cm^3. Use this to convert between volume and capacity.
- Volume is in cubic units (cm^3, m^3). Check unit consistency.

D1_Q6: Volume of Pyramids and Cones

Hook

The Great Pyramid of Giza contains over 2.3 million stone blocks. How do we calculate the space inside a pyramid? And what about an ice cream cone? Both are examples of shapes that taper to a point — their volume formulas are closely related to those of prisms and cylinders.

Prerequisites

- Volume of prisms and cylinders
- Area of circles and polygons
- Understanding of 3D geometry

Core Explanation

A **pyramid** has a polygonal base and triangular faces that meet at a point (the apex). A **cone** is like a pyramid with a circular base.

Volume Formulas

Shape	Formula
Pyramid (any base)	V = (1/3) x Area of base x Height
Rectangular pyramid	V = (1/3) x l x w x h
Cone	V = (1/3) x pi x r^2 x h

Key Insight

For any pyramid or cone, the volume is exactly **one-third** of the volume of the corresponding prism or cylinder with the same base area and height.

V(pyramid) = (1/3) x V(prism with same base and height)

Distinguishing Height from Slant Height

The **height** (h) is the perpendicular distance from the apex to the base centre.
The **slant height** (l) is the distance from the apex to the edge of the base along the sloping face.

Volume calculations use the perpendicular height, NOT the slant height.

Frustum
A frustum is a cone or pyramid with the top cut off parallel to the base.
V(frustum) = V(large cone) - V(small cone removed)

Worked Example

Problem: A rectangular pyramid has base length 8 cm, width 5 cm, and height 12 cm. Find its volume.

Step 1: Find base area. A = 8 x 5 = 40 cm^2

Step 2: Apply pyramid formula. V = (1/3) x 40 x 12 = (1/3) x 480

Step 3: Calculate. V = 160 cm^3

Answer: The volume is 160 cm^3.

Problem: A cone has radius 6 cm and height 14 cm. Find its volume (use pi = 3.142).

Step 1: V = (1/3) x pi x r^2 x h

Step 2: V = (1/3) x 3.142 x 36 x 14

Step 3: V = (1/3) x 3.142 x 504 = (1/3) x 1583.568

Step 4: V = 527.856 cm^3

Answer: The volume is approximately 527.86 cm^3.

Common Mistakes

X Forgetting the (1/3) factor and treating the pyramid like a prism.

OK Fix: Always include (1/3). A pyramid holds one-third the volume of a prism with the same base and height.

X Using the slant height instead of the perpendicular height.

OK Fix: The height must be measured perpendicular to the base. The slant height is longer. Use Pythagoras if n

X Using the diameter in the cone volume formula instead of the radius.

OK Fix: Cone volume uses radius. If given the diameter, halve it first: r = d/2.

Summary

- Volume of a pyramid = (1/3) x base area x height.
- Volume of a cone = (1/3) x pi x r^2 x h.
- Always use the perpendicular height, not the slant height.
- Empty/full ratio problems: use scale factor k (length ratio). Volume ratio = k^3.
- Frustum volume = large cone minus small cone.

D1_Q7: Volume of Spheres and Composite Solids

Hook

A soccer ball, a planet, a marble — spheres are all around us. Combined with other shapes like cylinders and cones, they form the composite solids we encounter in everyday objects. Understanding how to calculate the volume of spheres and composite solids is a key IGCSE skill.

Prerequisites

- Volume of prisms, cylinders, pyramids, and cones
- Area of circles
- Working with pi

Core Explanation

Volume of a Sphere

V = (4/3) x pi x r^3

Where r is the radius. The formula uses r^3 (cubed), not squared — volume is 3D.

Surface Area of a Sphere

SA = 4 x pi x r^2

Hemisphere

A hemisphere is half a sphere.
Volume of hemisphere = (2/3) x pi x r^3

Composite Solids

A **composite solid** is made up of two or more 3D shapes joined together (e.g., a cylinder with a hemisphere on top, or a cone on top of a cylinder).

Strategy:
1. Identify the simple solids that make up the composite shape.
2. Calculate the volume of each part separately.
3. Add or subtract as needed.

Sphere-Related Problems

- Melting and recasting: A sphere melted to form a cylinder. Volume is conserved.
- Floating/immersed objects: Volume of water displaced = volume of submerged object.
- Ratio problems: If radius doubles, volume increases by factor 8 (2^3).

Using Scale Factor

If two similar solids have a linear scale factor k:
- Areas scale by k^2
- Volumes scale by k^3

This is essential for comparing objects of different sizes.

Worked Example

Problem: A sphere has radius 6 cm. Find its volume (use pi = 3.142).

Step 1: V = (4/3) x pi x r^3 = (4/3) x 3.142 x 6 x 6 x 6

Step 2: V = (4/3) x 3.142 x 216 = (4/3) x 678.672

Step 3: V = 904.896 cm^3

Answer: The volume is approximately 904.90 cm^3.

Problem: A solid consists of a cylinder of radius 3 cm and height 8 cm with a hemisphere of the same radius on top. Find the total volume (use pi = 3.142).

Step 1: Volume of cylinder. V_cyl = pi x r^2 x h = 3.142 x 9 x 8 = 226.224 cm^3

Step 2: Volume of hemisphere. V_hemi = (2/3) x pi x r^3 = (2/3) x 3.142 x 27 = 56.556 cm^3

Step 3: Add. Total = 226.224 + 56.556 = 282.78 cm^3

Answer: The total volume is approximately 282.78 cm^3.

Common Mistakes

X Using r^2 instead of r^3 in the sphere volume formula.

OK Fix: Volume is 3D, so radius is cubed (r x r x r). Area formulas use r^2, but volume uses r^3.

X Using the diameter instead of the radius.

OK Fix: Always convert diameter to radius by halving it before using any sphere formula.

X Forgetting to include all parts of a composite solid.

OK Fix: Draw the shape, label each part (cylinder, cone, hemisphere, etc.), and calculate each separately befor

X Using the wrong factor: 4/3 pi r^3 for sphere vs 2/3 pi r^3 for hemisphere.

OK Fix: A hemisphere is exactly half a sphere: (1/2) x (4/3) x pi x r^3 = (2/3) x pi x r^3.

Summary

- Sphere volume: V = (4/3) x pi x r^3. Hemisphere: V = (2/3) x pi x r^3.
- Sphere surface area: SA = 4 x pi x r^2.
- For composite solids: calculate each part separately, then add or subtract.
- When a shape is melted and recast, volume is conserved.
- Scale factor k: volume scales by k^3, area scales by k^2.